

FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

Annual Examination 2026

Biology — Class IX

Syllabus: Ch 1 — The Science of Biology | Ch 2 — Biodiversity | Ch 3 — The Cell | Ch 4 — Cell Cycle
| Ch 5 — Tissues, Organs & Organ Systems | Ch 6 — Molecular Biology | Ch 7 — Metabolism (Sec
7.1-7.6)

ANSWER KEY (Version 2)

Total Marks: 65

Time Allowed: 2 Hours 30 Minutes

Version: 2

SECTION A — MCQ Answer Key (12 × 1 = 12 Marks)

Q1. C — Anatomy 1 mark

Anatomy is the study of internal body structures. **Physiology** studies functions; **Morphology** studies external size and shape; **Histology** studies tissues (not general internal structure).

Q2. C — Protista 1 mark

In Whittaker's five-kingdom system (1967), **Protista** includes single-celled eukaryotes. **Monera** contains prokaryotes; **Fungi** are multicellular eukaryotes with chitin; **Plantae** are multicellular photosynthetic eukaryotes.

Q3. B — can be crystallized but also reproduce 1 mark

Viruses show **non-living** traits (can be crystallized, no cellular structure, no enzyme system) and **living** traits (reproduce, contain genetic material, cause disease). They have no cell wall, do not respire, and are not bacteria.

Q4. C — Chitin 1 mark

Fungi cell walls are made of **chitin**. Plant cell walls use **cellulose**. Bacteria cell walls use **peptidoglycan**. **Pectin** is the middle lamella component found between plant cells.

Q5. B — S phase 1 mark

DNA replication occurs during the **Synthesis (S) phase** of interphase. G1 = growth and organelle production; G2 = spindle protein synthesis; M = mitotic division.

Q6. C — Urinary system 1 mark

The **urinary system** (kidneys, ureters, bladder, urethra) removes nitrogenous waste and regulates pH and water balance. The endocrine system secretes hormones; lymphatic defends against infection; digestive absorbs nutrients.

Q7. B — maximise light absorption for photosynthesis 1 mark

Palisade mesophyll cells are tightly packed in 2-3 cylindrical layers and contain many chloroplasts to capture maximum light. Gas diffusion occurs in the loosely arranged **spongy mesophyll**; xylem transports water; sclerenchyma provides support.

Q8. B — $C_x(H_2O)_y$ 1 mark

Carbohydrates are literally "hydrates of carbon" and their general formula is $C_x(H_2O)_y$, where x = carbon atoms and y = water molecules. For glucose: $C_6(H_2O)_6 = C_6H_{12}O_6$. Option D is only glucose, not the general formula.

Q9. C — Transcription 1 mark

Transcription is the first step of gene expression: DNA is copied into mRNA in the nucleus. **Translation** converts mRNA into protein at ribosomes. **Replication** copies the entire DNA. **Transduction** is gene transfer in bacteria.

Q10. C — Competitive inhibition 1 mark

Competitive inhibition is reversible: increasing substrate concentration displaces the inhibitor from the active site. Non-competitive inhibition binds the allosteric site and alters enzyme shape, so it cannot be reversed by adding more substrate.

Q11. B — Allosteric site 1 mark

Non-competitive inhibitors attach to the **allosteric site** (a point other than the active site), altering the enzyme's globular shape and damaging the active site. The prosthetic group is a permanently attached cofactor; the active site binds substrate.

Q12. C — break complex molecules into simpler ones 1 mark

Catabolism = breakdown reactions (complex → simple), releasing energy. Example: digestion, cellular respiration. **Anabolism** = building reactions (simple → complex). Photosynthesis and protein synthesis are anabolic. Energy is released in catabolism and utilized in anabolism.

SECTION B — Short Questions Answer Key (11 × 3 = 33 Marks)

2(i) Biology vs Biochemistry & Sub-fields of Biology 3 marks

Biology is the study of living organisms, while **Biochemistry** is the study of chemical compounds and processes in living organisms; it is a related science bridging biology and chemistry. 1 mark

Four sub-fields: 1 mark each — any 4 (award 2 for 4 correct)

- **Morphology** — studies external size, shape, and structure of organisms.
- **Physiology** — studies the functions and chemical processes of body parts.
- **Cytology** — studies the structure and function of cells.
- **Embryology** — studies the development of embryos from fertilization to birth.
- **Genetics** — studies heredity and variation in living organisms.
- **Ecology** — studies the relationship of organisms with their environment.

OR

THREE Careers in Biology 3 marks

- **Medicine/Surgery** — Degree: MBBS. Diagnoses and treats human diseases and disorders. 1
- **Biotechnology** — Studies organisms to develop useful products (vaccines, improved crops, medicine). 1
- **Agriculture/Farming** — Applies biological knowledge to improve crop yield and soil fertility for food production. 1

(Accept any 3 valid careers with degree/work: Fisheries, Animal Husbandry, Horticulture, Forestry.)

2(ii) Hypothesis vs Theory — Malaria Example 3 marks

A **hypothesis** is a proposed, testable explanation for an observation, based on limited evidence. A **theory** is a well-tested, evidence-supported explanation accepted by the scientific community. A hypothesis becomes a theory after repeated testing and confirmation. 1 mark

From the malaria case study:

Hypothesis — Laveran (1882) hypothesized that a microscopic parasite (later named *Plasmodium*) found in blood of malaria patients was the cause of the disease. 1 mark

Theory — Ronald Ross (1897) proved through experiments that the *Anopheles* mosquito transmits the *Plasmodium* parasite; this is now an accepted scientific theory about malaria transmission. 1 mark

OR

Laveran and Ronald Ross in Malaria Discovery 3 marks

Alphonse Laveran (1882): A French physician who examined blood of malaria patients under a microscope and discovered a parasite (now called *Plasmodium*). His hypothesis: a parasitic microorganism causes malaria. 1 mark

Ronald Ross (1897): A British physician who fed female *Anopheles* mosquitoes on malaria-infected patients, then dissected the mosquitoes and found *Plasmodium* in their stomach walls. His hypothesis: the *Anopheles* mosquito is the vector transmitting the parasite. He also proved the parasite passes into the next host when the mosquito bites. 2 marks

2(iii) Two-Kingdom vs Five-Kingdom Classification 3 marks

Two-kingdom system (proposed by **Aristotle/Linnaeus**): divided all living things into only **Plantae** (plants) and **Animalia** (animals). It was too simple and could not accommodate single-celled organisms, fungi, or bacteria properly. 1 mark

Five-kingdom system (proposed by **Robert Whittaker, 1967**): added three new kingdoms to Plantae and Animalia:

- **Monera** — prokaryotes (bacteria, cyanobacteria)
- **Protista** — single-celled eukaryotes
- **Fungi** — multicellular, absorptive heterotrophs with chitin cell walls

2 marks

OR

Domain Archaea 3 marks

Domain Archaea contains prokaryotes that live in extreme environments. They differ from bacteria in having no **peptidoglycan** in their cell walls, different membrane lipid structure, and different ribosomal RNA sequences (basis of Carl Woese's 1990 three-domain system). 1 mark

Three types:

- **Methanogens** — produce methane; found in swamps and animal intestines. 1
- **Halophiles** — live in highly salty environments (e.g., Dead Sea). 1
- **Thermoacidophiles** — live in hot acidic environments (e.g., volcanic hot springs).

They are placed in a separate domain because their rRNA sequences, membrane lipids, and antibiotic sensitivity differ fundamentally from Domain Bacteria.

2(iv) Golgi Apparatus — Structure and Function 3 marks

Discovered by **Camillo Golgi**. The Golgi apparatus consists of stacked, flattened membranous sacs called **cisternae**. 1 mark

Function: It receives proteins and lipids from the rough ER, modifies them (e.g., adds carbohydrate groups), packages them into **vesicles**, and directs them to their destinations inside or outside the cell.

1 mark

TWO products it forms:

- **Secretory vesicles** — carry proteins/enzymes to the cell surface for secretion.
- **Lysosomes** — contain digestive enzymes for intracellular digestion (autophagy).

1 mark

OR

Chloroplast vs Mitochondria 3 marks

Feature	Chloroplast	Mitochondria
Membrane	Double membrane; inner membrane forms thylakoids grouped into grana; fluid-filled stroma outside	Double membrane; inner membrane folds into cristae; fluid-filled matrix inside
Function	Photosynthesis — converts light energy into chemical energy (glucose)	ATP production — releases energy from glucose (aerobic respiration)
Specific reaction	Light-dependent reactions in thylakoids; Calvin cycle in stroma	Krebs cycle in matrix; ETC in inner membrane
Found in	Plant cells only	All eukaryotic cells

3 marks

2(v) Metaphase and Anaphase of Mitosis 3 marks

Metaphase: Chromosomes move to the **equatorial plate** (middle of the cell). Spindle fibres from both poles attach to the **centromere** of each chromosome, aligning them precisely. This is the stage where chromosomes are most visible and easiest to count. 1.5 marks

Anaphase: The **centromeres split**, separating each chromosome into two **chromatids**. Spindle fibres shorten and pull the chromatids to **opposite poles** of the cell. By the end of anaphase, each pole has a complete set of chromosomes. 1.5 marks

OR

G0 Phase and Cancer Cells 3 marks

G0 phase is a resting or quiescent state in which a cell exits the active cell cycle and stops dividing. Cells in G0 perform normal functions but do not replicate DNA. 1 mark

Two cell types entering G0:

- **Neurons (nerve cells)** — enter G0 permanently after development and never divide again.
- **Liver and kidney cells** — enter G0 temporarily but can re-enter the cycle if needed (e.g., after injury).

1 mark

Cancer cells do NOT enter G0 because the normal regulatory signals that tell a cell to stop dividing are disrupted. Cancer cells undergo **uncontrolled division**, bypassing G0, leading to tumour formation and eventually **metastasis** (spread to other tissues). 1 mark

2(vi) Prophase I of Meiosis 3 marks

Prophase I is the **longest phase of meiosis**. Events:

- Chromatin condenses into chromosomes and the nuclear membrane begins to break down.
- **Synapsis** — homologous chromosomes come together and pair up along their entire length. 1
- Each pair of homologous chromosomes forms a **bivalent** (tetrad of 4 chromatids: 2 chromosomes × 2 chromatids each). 1
- **Crossing over** — non-sister chromatids of homologous chromosomes exchange segments at points called **chiasmata**, creating new gene combinations and contributing to genetic variation. 1

OR

Significance of Meiosis 3 marks

Four significance of meiosis: 3 marks (any 4, award 3)

1. **Maintains chromosome number** — produces haploid (n) gametes so that when two gametes fuse at fertilization, the diploid (2n) number is restored in each generation.
2. **Genetic variation through crossing over** — exchange of DNA between homologous chromosomes in Prophase I creates new gene combinations.
3. **Genetic variation through independent assortment** — homologous pairs separate independently, producing many different chromosome combinations in daughter cells.
4. **Essential for sexual reproduction** — produces gametes (sperm and eggs) that carry half the genetic information of the parent.

2(vii) Homeostasis and Homeostatic Response 3 marks

Homeostasis is the tendency of living organisms to maintain a stable, relatively constant internal environment (e.g., temperature, pH, glucose level, ions). 1 mark

Homeostatic response sequence: 2 marks

Stimulus (change in internal environment) | Sensor (detects change – e.g., thermoreceptors in skin) | Control Centre (processes signal – e.g., Hypothalamus in brain) | Effector (responds – e.g., sweat glands, muscles, blood vessels) | Response (restores normal conditions) | Negative Feedback (response reverses the original change, returning system to set point)

OR

FOUR Types of Animal Tissues 3 marks

Tissue	Description	Example
Epithelial	Covers body surfaces and lines organs; forms skin and linings of organs	Skin epidermis, gut lining
Connective	Supports, connects, and protects body structures	Bone, tendon, fat (adipose)
Muscle	Produces movement by contraction (smooth, skeletal, cardiac)	Bicep (skeletal), heart wall (cardiac)
Nervous	Transmits electrical signals throughout body	Brain neurons, spinal cord

3 marks

2(viii) Shoot System vs Root System 3 marks

Shoot system: Located above the ground. Consists of the stem, leaves, and reproductive organs (flowers, fruits, seeds). Functions include **photosynthesis** (leaves), **transport** of food and water through xylem and phloem, and **reproduction**. 1.5 marks

Root system: Located below the ground. Consists of primary and secondary roots. Functions include **absorption of water and minerals** from soil, **anchoring** the plant, and **storage** of food (in some plants like carrot/potato). Roots lack chloroplasts so they do not photosynthesize. 1.5 marks

OR

Spongy Mesophyll and Gas Exchange 3 marks

Spongy mesophyll cells are loosely arranged with large **intercellular air spaces** between them. They are located beneath the palisade mesophyll layer in the leaf. 1 mark

Role in gas exchange: The large air spaces allow CO₂ and O₂ to diffuse freely between cells and exit/enter through **stomata** in the lower epidermis. 1 mark

Difference from palisade mesophyll:

- Palisade cells are tightly packed in 2-3 cylindrical layers with many chloroplasts for maximum light absorption.
- Spongy cells are irregular in shape, loosely arranged with fewer chloroplasts, optimized for gas diffusion rather than photosynthesis. 1 mark

2(ix) Saturated vs Unsaturated Fatty Acids 3 marks

Saturated fatty acids have all **single C-C bonds**, are **solid** at room temperature, and have the maximum number of hydrogen atoms attached.

Examples: cheese, butter, ghee, red meat. 1 mark

Unsaturated fatty acids have **double or triple C-C bonds** and are **liquid** at room temperature due to their kinked molecular structure.

Examples: soybean oil, mustard oil, olive oil, sunflower oil. 1 mark

One gram of lipid provides **9.1 kcal** of energy (double that of carbohydrates or proteins) because lipids have a higher carbon-hydrogen ratio. 1 mark

OR

FOUR Functions of Lipids + Energy Comparison 3 marks

Four functions of lipids: 2 marks (0.5 each)

- Energy storage** — lipids store long-term energy in fat cells and liver; 1 g gives 9.1 kcal.
- Cell membrane component** — phospholipids and cholesterol form the plasma membrane bilayer.
- Hormone synthesis** — steroid lipids form hormones like testosterone and aldosterone.
- Insulation** — lipids act as insulators against atmospheric heat and cold.

Why more energy: Lipids have a much higher **carbon-hydrogen ratio** than carbohydrates, so more C-H bonds are broken during oxidation, releasing more energy. 1 mark

2(x) Polysaccharides 3 marks

Polysaccharides are complex carbohydrates made of more than ten monosaccharide units. They are the largest group of carbohydrates and are usually **tasteless** (unlike mono- and disaccharides). 1 mark

Three polysaccharides and their roles: 2 marks

- **Starch** — stores extra glucose in plant roots, stems, fruits, and seeds; used as food by animals and humans.
- **Glycogen** — stored in animal muscle and liver cells as an energy reserve; quickly converted back to glucose when energy is needed.
- **Cellulose** — structural component of plant cell walls; provides strength and rigidity; acts as dietary fibre in humans, aiding digestion.

OR

Structure of a Nucleotide + Pentose Sugars 3 marks

A **nucleotide** is the monomer of nucleic acids (DNA and RNA). It has three components: 1 mark

1. **Pentose sugar** (5-carbon sugar) — the backbone of the nucleotide.
2. **Nitrogenous base** — a nitrogen-containing ring structure attached to carbon 1 of the pentose sugar.
3. **Phosphate group** — an acid group attached to carbon 5 of the pentose sugar.

Two types of pentose sugar: 2 marks

- **Deoxyribose** — found in **DNA**; lacks an -OH group at carbon 2.
- **Ribose** — found in **RNA**; has an -OH group at carbon 2.

2(xi) Effect of pH on Enzyme Activity 3 marks

pH affects enzyme activity by altering the **charge and shape** of the enzyme's active site. Each enzyme has an **optimum pH** at which it works best. 1 mark

- Slight change in pH above or below optimum **reduces** the rate of reaction.
- Extreme changes in pH cause **denaturation** of the enzyme (irreversible loss of shape).

1 mark

Example — Acidic: Pepsin (protein-digesting enzyme in stomach) works best at pH 2.0. **Example — Alkaline: Trypsin** (protein-digesting enzyme in small intestine) works best at pH 8.0. 1 mark

OR

THREE Types of Cofactors 3 marks

Many enzymes require a non-protein helper called a **cofactor**. Three types: 1 mark each

- **Activator** — detachable inorganic ions that activate the enzyme. Example: chloride ions (Cl^-) increase activity of salivary amylase; also Zn^{2+} , Fe^{2+} , Cu^{2+} .
- **Prosthetic group** — an organic molecule *permanently* and tightly attached to the enzyme. Example: Flavin Adenine Dinucleotide (FAD), Haem group.
- **Coenzyme** — a *detachable* organic cofactor that temporarily binds the enzyme. Example: NAD (nicotinamide adenine dinucleotide), Coenzyme A, Vitamins A and C.

SECTION C — Long Questions Answer Key (4 × 5 = 20 Marks)

3(i) The Biological Method — Six Steps with Malaria Example 5 marks

The **biological method** is a systematic process scientists use to investigate biological problems. Six steps in sequence: 4 marks (0.5 per step x 2 for step + malaria link)

1. **OBSERVATION** Scientists noticed that people living near swamps frequently developed fever and chills. Malaria: Laveran (1882) observed a parasite in blood of malaria patients under a microscope.
2. **PROBLEM IDENTIFICATION** A clear, focused question is framed. Malaria: "What causes malaria and how is it transmitted?"
3. **HYPOTHESIS** A testable explanation is proposed based on observation. Malaria: Laveran hypothesized that a microparasite (Plasmodium) causes malaria. Ross hypothesized that Anopheles mosquito transmits the parasite.
4. **DEDUCTION** Logical predictions are made: "IF the hypothesis is true, THEN..." "If mosquitoes transmit malaria, dissecting mosquitoes fed on infected patients should reveal the parasite in their bodies."
5. **EXPERIMENT** A controlled experiment tests the hypothesis. Ross: Fed female Anopheles mosquitoes on malaria patients, dissected them, and found Plasmodium in their stomach walls.
6. **CONCLUSION** Results either support or reject the hypothesis. Conclusion: Anopheles mosquito IS the vector; Plasmodium causes malaria.

Scientific Law vs Scientific Theory: 1 mark

A **scientific law** describes what happens in nature (a pattern or relationship), stated mathematically or verbally, without explaining why (e.g., Law of Biogenesis). A **scientific theory** explains why something happens; it is a well-tested, evidence-backed explanation (e.g., Cell Theory, Theory of Evolution).

OR

Three-Domain System of Classification (Carl Woese, 1990) 5 marks

Carl Woese proposed the **three-domain system** in 1990 based on differences in **ribosomal RNA (rRNA) nucleotide sequences**, membrane lipid structure, and antibiotic sensitivity. 1 mark

Domain Bacteria: Prokaryotes with **peptidoglycan** cell walls. Found in most habitats. Examples: *E. coli*, *Bacillus*. 1 mark

Domain Archaea: Prokaryotes WITHOUT peptidoglycan; live in extreme environments. Types: methanogens (swamps), halophiles (salty water), thermoacidophiles (hot springs). 1 mark

Domain Eukarya: All eukaryotic organisms. Contains four kingdoms: 2 marks

Kingdom	Cell type	Nutrition	Example
Protista	Unicellular eukaryote	Varied (autotrophic/heterotrophic)	Amoeba, Paramecium
Fungi	Multicellular (mostly)	Absorptive heterotroph	Mushroom, yeast
Plantae	Multicellular	Autotrophic (photosynthesis)	Mango tree, rose
Animalia	Multicellular	Heterotrophic (ingestive)	Human, frog

3(ii) Structure and Functions of the Nucleus 5 marks

The nucleus is the **control centre** of the cell, containing the genetic information (DNA) and directing all cellular activities. 1 mark

Component	Structure and Function
Nuclear Envelope	Double membrane surrounding the nucleus; outer membrane is continuous with rough ER. Controls entry/exit of materials. 1
Nuclear Pores	Small openings in the nuclear envelope that allow mRNA, ribosomes, and other molecules to move between nucleus and cytoplasm.
Nucleoplasm	Fluid matrix inside the nucleus in which chromatin and nucleolus are suspended.
Chromatin	Thin fibres of DNA wrapped around histone proteins. Chromatin = DNA + histones. In a non-dividing cell, DNA is in this loosely coiled form. 1
Nucleolus	Dense body inside nucleus; site of rRNA synthesis, which combines with proteins to form ribosomes. 1

Chromatin to Chromosomes: When the cell prepares to divide, chromatin fibres condense and coil further, forming thread-like **chromosomes**. Each chromosome consists of DNA coiled around **histones** to form bead-like **nucleosomes**. Further coiling packs DNA tightly. A small portion of a chromosome is called a **gene**, which carries the code for one protein. The complete set of genes in an organism is called its **genome**. 1 mark

OR

Complete Process of Meiosis I and II 5 marks

Meiosis I (Reduction Division): 2 marks

- **Prophase I** (longest): Chromosomes condense, synapsis occurs forming bivalents, crossing over at chiasmata.
- **Metaphase I**: Bivalents align at the equatorial plate; spindle fibres attach to centromeres of each homologue.
- **Anaphase I**: Homologous chromosomes (not chromatids) separate and are pulled to opposite poles.
- **Telophase I + Cytokinesis**: Two haploid cells form, each with one chromosome from each homologous pair.

Meiosis II (Similar to Mitosis): 2 marks

- **Prophase II**: Chromosomes condense again; spindle forms.
- **Metaphase II**: Chromosomes align at equator.
- **Anaphase II**: Centromeres split; sister chromatids pulled to opposite poles.
- **Telophase II + Cytokinesis**: Four haploid (n) daughter cells produced.

Meiosis vs Mitosis: 1 mark

Feature	Mitosis	Meiosis
Divisions	1	2
Daughter cells	2 diploid (2n)	4 haploid (n)
Genetic outcome	Genetically identical (clones)	Genetically variable (crossing over)
Purpose	Growth, repair, asexual reproduction	Sexual reproduction (gamete production)

3(iii) Watson and Crick Model of DNA 5 marks

DNA (Deoxyribonucleic Acid) is a **double-stranded polynucleotide** macromolecule. Its structure was explained by **James Watson and Francis Crick**. 1 mark

Seven key points of the Watson-Crick model: 3 marks

1. The DNA molecule is **double stranded**, shaped like a **ladder coiled into a helix** (double helix).
2. Both strands are **antiparallel** (run in opposite 5'→3' directions to each other).
3. The **backbone (uprights)** of the ladder is made of alternating **sugar (deoxyribose)** and **phosphate** groups.
4. The **rungs** of the ladder are formed by pairs of **nitrogenous bases** held together by hydrogen bonds.
5. **Adenine (A) always pairs with Thymine (T)** via 2 hydrogen bonds; **Cytosine (C) always pairs with Guanine (G)** via 3 hydrogen bonds. (Chargaff's rule: A=T, C=G.)
6. The sequence of nucleotides in one strand is **complementary** to the sequence in the other strand.
7. The diameter of the double helix is **uniform at ~2 nm**.

Gene and Genome: A **gene** is a specific localized region of DNA on a chromosome that codes for a particular protein. The total number of genes found in the complete set of chromosomes of an organism is called its **genome**. 1 mark

OR

Classification of Carbohydrates 5 marks

Carbohydrates are classified on the basis of the number of **saccharide (monosaccharide) units** in the molecule. General formula: $C_x(H_2O)_y$; ratio of C:H:O = 1:2:1. 1 mark

Type	Definition	Molecular Formula	Examples
Monosaccharides	Simplest carbohydrates; single saccharide unit; cannot be hydrolysed further; sweet, crystalline, soluble in water	Glucose: $C_6H_{12}O_6$ Ribose: $C_5H_{10}O_5$	Glucose, Fructose, Galactose, Ribose
Disaccharides	Two monosaccharides bonded by condensation (releasing water); slightly less sweet	$C_{12}H_{22}O_{11}$	Sucrose, Maltose, Lactose
Polysaccharides	More than 10 monosaccharides; complex, tasteless, storage or structural carbohydrates	$(C_6H_{10}O_5)_n$	Starch, Glycogen, Cellulose, Chitin

2 marks

Formation of Disaccharides: 2 marks

- **Sucrose** (cane sugar) = Glucose + Fructose → $C_{12}H_{22}O_{11}$ + H_2O
- **Maltose** (malt sugar) = Glucose + Glucose → $C_{12}H_{22}O_{11}$ + H_2O
Produced during hydrolysis of starch and glycogen; found in beer brewing.
- **Lactose** (milk sugar) = Galactose + Glucose → $C_{12}H_{22}O_{11}$ + H_2O
Found in milk of all mammals (2-8%); can be prepared from whey (cheese by-product).

3(iv) Characteristics of Enzymes — Six Characteristics 5 marks

Enzymes are biologically active **globular proteins** made by living cells that tremendously speed up biochemical reactions without being consumed. 0.5 mark

Six characteristics: 3 marks (0.5 each)

1. **Enzymes are globular proteins** — made of chains of amino acids folded into 3D globular protein molecules. A few enzymes are RNA, called ribozymes.
2. **Enzymes increase rate of reaction** — speed up reactions millions of times faster than non-enzymatic reactions. They do NOT start reactions, only speed them up.

3. **Enzymes are required in small quantities** — their structure is slightly changed during reaction but regains original shape at end, so they can be reused indefinitely.
4. **Enzymes are specific** — each enzyme catalyses only one type of reaction and acts on only one type of substrate (e.g., amylase acts on starch only, not protein or fats).
5. **Enzymes require cofactors** — many enzymes need a non-protein component (cofactor) for proper functioning. Types: activator (inorganic ions), prosthetic group (permanently attached organic), coenzyme (detachable organic).
6. **Enzymes regulate production and activity** — enzyme synthesis can be increased or decreased by cells as needed; inhibitors and activators regulate enzyme activity.
7. **Enzymes make metabolic pathways** — many enzymes work in sequence, each acting on the product of the previous reaction to build a metabolic pathway.

(Any 6 of the 7 above accepted.)

Activation Energy: The minimum energy needed to start a reaction is called **activation energy**. Enzymes **lower the activation energy** by binding to the substrate and providing an alternative reaction pathway, allowing reactions to occur at body temperature. **1 mark**

Metabolic Pathway: A series of connected chemical reactions where multiple intermediate molecules are produced before an initial substrate is converted to the final product. Each step is catalysed by a specific enzyme. **0.5 mark**

OR

Effect of Substrate Concentration and Temperature on Enzyme Activity **5 marks**

Effect of Temperature: **2 marks**

As temperature increases from low to optimum, enzyme activity increases because molecules move faster, increasing collision frequency between enzyme and substrate.

- **Optimum temperature** for human enzymes: 36°C–38°C (body temperature). Enzyme works fastest here.
- Above optimum: atoms of enzyme vibrate vigorously, globular structure and active site are damaged. This is called **denaturation**. Activity falls sharply to zero.
- At very low temperature: enzyme is inactive due to insufficient activation energy; activity resumes when normal temperature is restored.

Rate of Reaction | / \ | / \ (denaturation) | / \ | / \ | ____ / ____ \ ____ Temperature (°C) ^
Optimum temp (~37°C for human enzymes)

Effect of Substrate Concentration: **2 marks**

When enzyme molecules are in excess, increasing substrate concentration increases reaction rate because more substrate molecules fill the available active sites.

- Up to a certain point, rate is directly proportional to substrate concentration.
- Beyond a maximum: all active sites are occupied — called **saturation of active sites**. Further addition of substrate has NO effect on rate.

Rate of ----- (saturation: all active sites occupied) Reaction / / / /
_____/_____ Substrate Concentration

Definitions: **1 mark**

- **Denaturation:** Irreversible loss of an enzyme's 3D structure and active site shape due to extreme temperature or pH.
- **Optimum temperature:** The specific temperature at which an enzyme shows maximum catalytic activity.
- **Saturation of active site:** The condition when all available enzyme active sites are occupied by substrate molecules, so no further increase in reaction rate occurs even with more substrate.